An Emerging Immersive Technology-A Survey

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***Abstract*- Immersive technology refers to technology that attempts to emulate a physical world through the means of a digital or simulated world by creating a surrounding sensory feeling, thereby creating a sense of immersion. Immersive technology enables mixed reality; which is a combination of Virtual reality and augmented reality or a combination of physical and digital. Immersive technology is applied in several areas, including retail and e-commerce, the adult industry, art, entertainment and videogames and interactive storytelling, military, education, and medicine. In this article we have discuss about the uses of Immersive Technology, Realties, devices along with the advantages and disadvantages in it. Also it is explained with the Virtual and mixed reality, future of immersive technology in different fields and its innovations.**

***Index terms*- Immersive Technology, Augmented Reality, Virtual Reality, Mixed Reality, AR hologram, VR suit**

1. INTRODUCTION

Immersive Technology is an integration of virtual content with the physical environment in a way that allows the user to engage naturally with the blended reality. The definition of immersion is wide and variable but here it is assumed to mean simply that the user feels like they are part of the simulated "universe". An immersive digital environment could be a model of reality, but it could also be a complete fantasy user interface or abstraction, as long as the user of the environment is immersed within it. The success with which an immersive digital environment can actually immerse the user is dependent on many factors such as believable 3D computer graphics, surround sound etc… New technologies are currently under development which claims to bring realistic environmental effects to the player's environment effects like wind, seat vibration and ambient lightning.

2. USES OF IMMERSIVE TECHNOLOGY

In today's world immersive plays an important role in everywhere. It is applied in several areas, including retail and e-commerce, the adult industry, art, entertainment and video games and interactive storytelling, military, education and medicine. There are several ways to use Immersive Technology.

2.1 Exploration of Physical Locations

Travel brands can use Immersive Technology to help those who are afraid of riding an airplane or even a boat by using the technology to immerse them in an environment that would calm them down or allow them to think of something else.

2.2 Marketing & Advertising

2.2.1 Immersive Advertising

Immersive Advertising is an entirely new ad format that leverages Immersive Technology while still being able to use today's current ad channels and ecosystems. This new ad technology takes advantage of a mobile phone's sensory inputs (like its gyroscope, magnetometer, and accelerometer). These sensory inputs allow the immersive ad to respond to device motion, which makes it very effective for mobile advertising.

2.2.2 Experiential Marketing

Experiential marketing is one of the most common implementations of Immersive Technology in business where brands can set up interactive booths in trade shows, country fairs, malls or even sporting events where people can try promotional VR experiences. These promotional experiences can be based on an upcoming movie or show or they can even be a promotional experience for a product.

2.2.3 Immersive Web Apps

Immersive Ads work best in mobile and it can be flexible as it works either in mobile web or in-app. Immersive Ads allow consumers to explore a brand within an ad unit seamlessly while browsing the

internet-all without having to be redirected into a page. Immersive Web App (IWA) is a bigger and deeper implementation of an immersive advertisement.

2.2.4 Ecommerce and Shopping Tools

Immersive Technology has put new meaning to the term '"online shopping." There are more stores are using AR apps to allow users to test out their products beforehand. Even with VR, brands like those within the car industry are using virtual showrooms where consumers can look at cars, take virtual test drives and even "sit inside" them. The experience is made even more immersive by letting users sit in actual car seats to make them feel like they're actually inside a car. NOTE: In marketing & advertising, Immersive Technology allows brands to reach consumers in every stage of the consumer purchasing funnel.

2.3 Healthcare

Doctors and healthcare providers have discovered that Immersive Technology helps with relieving a patient's pain as immersing them in an experience distracts them from what's going on. For example, in a university in Canada, students developed a VR game that would immerse cancer patients during chemotherapy and thus, distract them from their pain. Another benefit (which ties in with Education) is that Immersive Technology allows medical students and doctors to do surgery simulations. Immersive Technology can help doctors watch other surgeons perform operations remotely-maybe even from a first person perspective. It improves the healthcare sector because it is a really good tool that promotes empathy as well. Doctors and healthcare providers would better understand their patients by immersing them in experiences that allow them to walk in their patient's shoes and see what they experience.

2.4 Education

Education has and can greatly benefit from Immersive Technology as it taps into the different senses of a human being. Existing multimedia systems used in education mostly address only two senses by using two communication channels (audio & visual) of the 5 human senses limiting the potential efficiency of learning. At the same time modern immersive technologies enable to employ not only visual and audio but also olfactory and haptic media. Immersive Technology like AR or even MR can even help students learn about the human body and how each part functions by overlaying digital images on an actual person and highlighting where the different parts are located.

2.5 Art

Many museums are finding that using Immersive Technologies allows visitors to engage with pieces and "feel a connection with the artist". By revitalizing old works of art in the museum into virtual spaces, visitors no longer have to stand in line among large crowds in order to stare at a painting from 5 feet away. Now they can be immersed within a classic painters studio or find themselves placed in the landscape of the painting. And with the ability to download immersive technology apps, those who are unable to travel to popular museums are now given an opportunity to view rare works of art and limited run collections.

2.6 Gaming

The gaming industry has been one of the first movers of Immersive Technology. Because of this, Gaming and Immersive Technology have become almost synonymous to each other-causing a barrier for brands to leverage this technology. The launch of the Play station VR and even Pokemon Go has definitely helped in this aspect.

Immersive Technology is here to stay and is ushering in the next Industrial Revolution. This new technology is opening up a realm of possibilities that the world has never seen before and is making improvements in industries that have not had a major technological shift in a long time.

3. TRENDING REALITIES IN IMMERSIVE TECHNOLOGY

1. VIRTUAL REALITY (VR)

Virtual Reality is a simulated experience that can be similar to or completely different from the real world. Applications include virtual reality can include entertainment purposes. Virtual reality's roots lie in a combination of 3-D still images, computer gaming, computer-assisted instruction, equipment simulators etc Some VR setups display immersive environments

through head-mounted displays that replace or supplement your view of the real world.

1.1 Uses of Virtual Reality

A technology first prototyped in the 1970's virtual reality simulates a real or imagined physical event, such as a walk through the park, or on mars. Modern VR typically produces artificial scenes in software and renders them as an immersive experience of sight and sound. VR has found many uses in education, entertainment, medicine and science.

1.1.1 VR Hardware

The hardware for VR begins with a computer or console capable of high-speed graphics processing; high performance is essential to VR because generating realistic, moving scenery involves lots of math. To view the graphics, VR system users wear a special set of goggles or a closed, enveloping helmet that incorporates a wide-angle display. In addition to displaying imagery, the device tracks head movement; when you turn or tilt your head, you can see what's behind, above or below you in the virtual world. Other VR setups employ a wraparound screen in a small enclosed room or chamber in which you sit. In either case, the VR system creates a detailed visual panorama.

1.1.2 Computer Games

VR adds new dimensions of realism and engagement to computer games; the virtual world of the game surrounds you, allowing you to explore it and interact with the characters. The characters may be avatars—graphical stand-ins—used by other players or may be entirely fictional characters with computer-generated personalities. Many VR games handle movement and action with a traditional joystick or game controller.

1.1.3. Health Care

People suffering from phobias and post-traumatic stress disorder have in some cases found relief through VR-based treatments; the system puts the patient in a series of progressively challenging emotional encounters, helping him gradually cope with anxiety-inducing situations. Also using VR, surgeons can explore and plan complex medical procedures by evaluating virtual patients created from CT scans.

1.1.4 Vehicle Simulators

Flight simulators are a classic use of VR; the simulator is an enclosed room built on a mechanical platform that reproduces the motions encountered during flight. In the room, pilots manipulate realistic controls while responding to a variety of flying conditions. In much the same way, tractor- trailer and other commercial and construction truck simulators, improving safety and driver confidence

Advantages & Disadvantages of Computerized Systems

Advantages & Disadvantages of computerized systems will depend on user's own experience. For those who use computerized systems both at home and at work, you may even find that you have opposite perspectives on whether advantages outweigh disadvantages or the other way around when it comes to comparing your home computerized system with your office's system.

1.2ADVANTAGE

Efficiency

Computerized systems allow for greater efficiency in performing specific tasks both more accurately and more rapidly than doing the same task using, for example, a pencil and paper or a pencil and ledger book. Computerized systems also offer storage capacity efficiency. For example, a computerized systems hard drive taking up a relatively small amount of space can store a large collection of business or paper documents. A paper-based system, in contrast, would take up at least one, and usually more, file cabinets for storage of the same document collection.

Learning

Computerized systems can also enliven young people's learning activities, as well as and accommodate a variety of different learning styles. In an office setting, computer training, or CBT, allows management training flexibility. At the most fundamental level, CBT learning modules incorporate only slide-like pages of text and perhaps some simple questions to test comprehension. At the most sophisticated level, CBT training can incorporate simulations of complex situations that the learner may encounter in her profession. Both the aviation and the medical professions make use of computer-based simulations in some of their training

activities. Management also has the flexibility to combine CBT instructional activities with more traditional forms of training, such as instructor-led classroom training.

1.3 DISADVANTAGE

Miscellaneous

Rapid changes in technology can cause a computer to become out dated relatively quickly, requiring upgrades on a regular basis for computerized systems deployed in business use. Computerized systems can also give some users a sense of false security that can result in catastrophic losses of massive amounts of present or past work unless the user learns to use proper backup procedures. Computer work also has been implicated in a variety of health-related issues, such as skeletal problems, eyestrain and a complex of conditions known as repetitive stress injury-with carpal tunnel syndrome illustrating the best-known repetitive-stress condition.

Social

Socially, children who spend excessive time using computers have less opportunity to learn such social skills as good manners or cooperation with others. Adults whose work involves heavy computer use to the extent of restricting social interaction may feel isolated or cut off from the world. While home-based workers may suffer from the greatest sense of isolation, this can affect certain employees of traditional business, also. A programmer or a quality tester who works alone in a lab or back office environment can also experience such a sense of seclusion.

Workplace Concerns

Computerized systems exemplify both advantages and disadvantages in the workplace. The use of computerized systems has resulted in streamlining such that, for example, the same size organization needs fewer secretaries for the same number of executives or departments than it on once did. This loss of many lower-and middle-level jobs, combined with the rapid changes in technology that can fairly rapidly outdate even some jobs within the rapid changes in technology that can fairly rapidly outdate even some jobs within the computer or information technology field itself, has given rise to a perception of lack of job security. Conversely, the prevalence of computerized systems has also resulted in the creation of challenging new jobs, sometimes opening up entirely new professional fields.

2. AUGMENTED REALITY (AR)

2.1 What is Augmented Reality?

Augmented reality is the technology that expands our physical world, adding layers of digital information onto it. Unlike Virtual Reality (VR), AR does not create the whole artificial environments to replace real with a virtual one. AR appears in direct view of an existing environment and adds sounds, videos, and graphics to it. A view of the physical real-world environment with superimposed computer-generated images, thus changing the perception of reality, is the AR. The term itself was coined back in 1990, and one of the first commercial uses was in television and military. With the rise of the Internet and smart phones, AR rolled out its second wave and nowadays is mostly related to the interactive concept. 3D models are directly projected onto physical things or fused together in real-time, various augmented reality apps impact our habits, social life, and the entertainment industry. AR apps typically connect digital animation to a special ‘marker', or with the help of GPS in phones pinpoint the location. Augmentation is happening in real time and within the context of the environment, for example, overlaying scores to live feed sport events.

2.2 Brief history of AR

2.2.1 AR in the 1960s: In 1968 Ivan Sutherland and Bob Sproull created a first head-mounted display; they called it The Sword of Damocles. Obviously, it was a rough device that displayed primitive computer graphics.

2.2.2 AR in the 1970s: In 1975 Myron Krueger created Video place – an artificial reality laboratory. The scientist envisioned the interaction with digital stuff by human movements. This concept later was used for certain projectors, video cameras, and onscreen silhouettes.

2.2.3 AR in the 1980s: In 1980 Steve Mann developed a first portable computer called Eye Tap, designed to be worn in front of the eye. It recorded the scene to superimposed effects on it later, and show it all to a user who could also play with it via

head movements. In 1987 Douglas George and Robert Morris developed the prototype of a heads-up display (HUD). It displayed astronomical data over the real sky.

2.2.4 AR in the 1990s: The year 1990 marked the birth of the "augmented reality" term. It first appeared in the work of Thomas Caudell and David Mizell – Boeing company researchers. In 1992 Louis Rosenberg of the US Air Force created the AR system called "Virtual Fixtures". In 1999, a group of scientists led by Frank Delgado and Mike Abernathy tested new navigation software, which generated runways and streets data from a helicopter video.

2.2.5 AR in the 2000s: In 2000 a Japanese scientist Hirokazu Kato developed and published AR Toolkit – an open-source SDK. Later it was adjusted to work with Adobe. In 2004 Trimble Navigation presented an outdoor helmet-mounted AR system. In 2008 Wikitude made the AR Travel Guide for Android mobile devices.

2.2.6 AR today: In 2013 Google beta tested the Google Glass – with internet connection via Bluetooth. In 2015 Microsoft presented two brand new technologies: Windows Holographic and HoloLens (an AR goggles with lots of sensors to display HD holograms). In 2016 Niantic launched Pokemon Go game for mobile devices. The app blew the gaming industry up and earned $2 million in a just first week.

2.3 DEVICES:

AR can be displayed on various devices: Screens, glasses, handheld devices, mobile phones, head-mounted displays. It involves technologies like S.L.A.M. (simultaneous localization and mapping), depth tracking (briefly, a sensor data calculating the distance to the objects), and the following components:

2.3.1 Cameras and sensors: Collecting data about user's interactions and sending it for processing. Cameras on devices are scanning the surroundings and with this info, a device locates physical objects and generates 3D models. It may be special duty cameras, like in Microsoft hololens, or common smartphone cameras to take pictures/videos.

2.3.2 Processing: AR devices eventually should act like little computers, something modern smartphones already do. In the same manner, they require a CPU, a GPU, flash memory, RAM, a GPS, etc. to be able to measure speed, angle, direction, orientation in space, and so on.

2.3.3 Projection: This refers to a miniature projector on AR headsets, which takes data from sensors and projects digital content (result of processing) onto a surface to view. In fact, the use of projections in AR has not been fully invented yet to use it in commercial products or services.

2.3.4 Reflection: Some AR devices have mirrors to assist human eyes to view virtual images. Some have an "array of small curved mirrors" and some have a double-sided mirror to reflect light to a camera and to a user's eye. The goal of such reflection paths is to perform a proper image alignment.

2.4 TYPES OF AUGMENTED REALITY:

There are 4 types of augmented reality

1. Marker-based AR:

Some also call it to image recognition, as it requires a special visual object and a camera to scan it. It may be anything, from a printed QR code to special signs. The AR device also calculates the position and orientation of a marker to position the content, in some cases. Thus, a marker initiates digital animations for users to view, and so images in a magazine may turn into 3D models.

2. Marker less AR:

A.k.a. location-based or position-based augmented reality that utilizes a GPS, a compass, a gyroscope, and an accelerometer to provide data based on user's location. This data then determines what AR content you find or get in a certain area. With the availability of smart phones this type of AR typically produces maps and directions, nearby businesses info. Applications include events and information, business ads pop-ups, navigation support.

3. Projection-based AR:

Projecting synthetic light to physical surfaces, and in some cases allows to interact with it. These are the holograms we have all seen in sci-fi movies like Star Wars. It detects user interaction with a projection by its alterations.

4. Superimposition-based AR:
Replaces the original view with an augmented, fully or partially. Object recognition plays a key role, without it the whole concept is simply impossible. We've all seen the example of superimposed augmented reality in IKEA catalogapp that allows users to place virtual items of their furniture catalog in their rooms.

2.5 AUGMENTED REALITY DEVICES:
Many modern devices already support augmented reality. From smartphones and tablets to gadgets like Google Glass or handheld devices, and these technologies continue to evolve. For processing and projection, AR devices and hardware, first of all, have requirements such as sensors, cameras, accelerometer, gyroscope, digital compass, GPS, CPU, displays, and things we've already mentioned. Devices suitable for augmented reality fall into the following categories:
2.5.1 Mobile devices (smartphones and tablets) – the most available and best fit for AR mobile apps, ranging from pure gaming and entertainment to business analytics, sports, and social networking. Special AR devices, designed primarily and solely for augmented reality experiences. One example is head-up displays (HUD), sending data to a transparent display directly into user's view. Originally introduced to train military fighter's pilots, now such devices have applications in aviation, automotive industry, manufacturing, sports, etc.

2.5.2 AR glasses (or smart glasses) – Google Glasses, Meta 2 Glasses, Laster See-Thru, Laforge AR eyewear, etc. These units are capable of displaying notifications from your smartphone, assisting assembly line workers, access content hands-free, etc.

2.5.3 AR contact lenses (or smart lenses), taking Augmented Reality one step even farther. Manufacturers like Samsung and Sony have announced the development of AR lenses. Respectively, Samsung is working on lenses as the accessory to smartphones, while Sony is designing lenses as separate AR devices (with features like taking photos or storing data).

2.5.4 Virtual retinal displays (VRD), creating images by projecting laser light into the human eye. Aiming at bright, high contrast and high-resolution images, such systems yet remain to be made for a practical us.

2.6 POSSIBLE APPLICATIONS OF AR:
Augmented reality may complement our everyday activities in various ways. For instance, one of the most popular applications of AR is gaming. New AR games provide much better experiences to players, some even promote a more active outgoing way of life (PokemonGo, Ingress). Gaming grounds are being moved from virtual spheres to real life, and players actually perform certain activities. For instance, a simple gym activity for kids by the Canadian company SAGA, where to crack cubes moving on wall children hit it with a ball. AR in retail may act to bring better customer engagement and retention, as well as brand awareness and more sales. Some features may also help customers make wiser purchases – providing product data with 3D models of any size or color. Real-estate can also benefit from Augmented Reality via 3D tours of apartments and houses that can also be manipulated to amend some parts.

Other potential areas for AR include:
Education: Interactive models for learning and training purposes, from mathematics to chemistry.
Medicine/healthcare: To help diagnose, monitor, train, localize, etc.
Military: For advanced navigation, marking objects in real time.
Art / installations / visual arts / music. Tourism: Data on destinations, sightseeing objects, navigation, and directions.
Broadcasting: Enhancing live events and event streaming by overlaying content.
Industrial design: To visualize, calculate or model.

3. MIXED REALITY
3.1 Why mixed reality has an edge over virtual reality
Virtual reality transposes users to an alternate world by offering interactive capabilities. Whereas, Mixed Reality with the blend of VR and AR changes the way users create, connect, and collaborate with a new holographic experience. Therefore, MR is the better choice for industries that are trying to improve their operational efficiency using holographic experiences. Nowadays, many enterprises want to leverage various

types of 'immersive experiences', including augmented reality, virtual reality, and mixed reality. Out of these three, mixed reality (MR) has become the common goal for many enterprises due to the distinctive capabilities compared to other immersive experiences. MR is capable of producing enterprise-focused applications with the best combination of both physical and digital worlds. This is why many enterprises are investing in MR applications development to deliver a digital experience to their users. The main mixed reality devices available in the market are Microsoft HoloLens and ODG headsets that help different industries improve the way they work.

3.2 How mixed reality works

Mixed reality (MR) is the mix of real and virtual worlds where physical and digital objects co-exist and interact in real time. MR comes with various powerful features such as mapping physical surroundings, monitoring gestures, and language processing for voice recognition and more. MR indicates a new way of working by communicating and collaborating with its practical uses with enterprises, as it offers 'real-world, real-life' experiences.

3.3 How Mixed Reality is better than Virtual Reality

In comparison, virtual reality has its limitations as it completely blocks the user's surroundings and thus, requires the user to remain in place. A few VR headsets allow for limited mobility but not as much when compared to a mixed reality device such as HoloLens. MR relates to the user's present surroundings with 3D visuals and creates a continuous view of both worlds; virtual and real. Also, with MR technology the user is allowed to have movements and remain productive while attending tasks. This is one of the main advantages of MR over VR, where MR has more impact on the enterprise market although VR is on the development curve. However, challenges involving real-time processing and calibration of physical environments need to be overcome. Enterprises adopting MR are likely to be highly effective, the reason being it will provide a rich experience and personalized views without any distractions.

3.4 Let's see how MR can benefit various industries:

3.4.1 Manufacturing:

The industrial environment can greatly improve operational efficiencies from assembly lines to supply chains using mixed reality head-worn display. With MR devices, manufacturers can perform product designing, product maintenance, quality inspection, and more by collaborating in real-time with different teams across the organization. Even, the mixed reality solution is perfect for enabling rapid prototyping, 3D model visualization, and quick decision-making.

3.4.2 Healthcare:

MR has many wide applications and features to offer the healthcare industry such as the 3D operating room with simulations, broadcasting options, virtual organ models for complex surgeries and more. This is a powerful way of transforming knowledge and skills with combined data analytics, sensors and artificial intelligence (AI), to practice evidence-based medicine for delivering care.

3.4.3 Retail:

In retail, the stores that use MR are able to offer new features of real experience to their clients such as checking out visual designs of items that mix and match, moving around some of the objects superimposed in real time. One of the best examples is the Lowe's retailers, they used HoloLens headset to allow their clients build a kitchen of their choice using generic showroom template.

3.4.4 Real Estate:

Property builders and realtors can provide their clients immersive 360o virtual walkthrough of the property under development or in the planning phase using mixed reality. Moreover, clients can also decide on furniture designs, colors of the walls and tiles and for more customization. This way MR can allow them to make a better decision. To conclude, mixed reality technology is still new and emerging, and it is therefore important for enterprises to consider primary advantages as of now. Many use cases, interactive ways, applications and even hardware of mixed reality will continue to evolve with the rise of this technology. This is hence the suitable opportunity for enterprises to invest in this technology. MR has the potential to change how people work, communicate and relate to the world.

4. FUTURE OF IMMERSIVE TECHNOLOGY

Digital transformation will use data and emerging technologies to create powerful new experiences and will strengthen businesses and allow them to remain competitive. Technology is immersed in our lives and our businesses more than ever: it is increasingly pervasive in the way we work and lead our organisations. At the heart of this is data, which organisations have been stockpiling for years. By 2020, this data volume will reach 44 trillion gigabytes or 44 zettabytes, and organisations are already working hard to extract greater value from that data. What are driving this is experiences: of being an employee, or a customer, or a business leader. The importance of data-derived insights in transforming experiences – within the workplace and outside – will continue to grow and is set to define many leaders' priorities as they work towards creating digital-first organisations.

4.1 Technology in the workplace

Why is experience a driver? Look at the expectations of the workforce today. Generation (Gen) Z, the first true digital-natives who are just starting to enter the workforce, provides a timely example. According to our latest study, Gen Z: The future has arrived, 98% have already used technology as a part of their formal education, and 85% want to work with cutting-edge technology in their future careers. For this generation, technology is an expectation, not a new factor in their lives or work. Organizations are being challenged to think about how they can attract the best talent, by re-examining the entire experience they are offering their workforce: from learning to a commitment to accelerating career paths, to genuine enablement through technology. These are strategies that have a direct impact on business outcomes. To the latter point, immersive intelligence will continue to transform the workforce experience: half of the business leaders expect to learn on the job with augmented reality (AR) by 2030. The PCs and devices we use every day will continue to learn from our habits, proactively booting-up with the right apps and services at the right time. Advances in natural language processing and voice technologies will create a more productive dialogue with machines, while automation and robotics will create faster, more fluid collaboration with technology to get more done. With augmented and virtual reality (VR) applications creating workplace and home experiences, people will have access to the data they need to do work whenever, wherever they are.

4.2 Customer engagement

Immersive experiences are quickly becoming a focal point for companies' customer engagement strategies, too. Globally, 8 in 10 business leaders expect that they will be delivering hyper-connected customer experiences with VR, by 2030, according to our research. By combining machine intelligence with the latest augmented and virtual reality solutions, providers of consumer technology are making virtual assistants more intelligent and accessible than ever before. With the rising number of smart home technologies, "things" and connected cars – all of which can learn our preferences and proactively serve-up targeted content and information based on previous interactions – immersive intelligence will be increasingly important in terms of its pervasiveness and its impact on the way we work, live, and play. From enabling machine automation in the workplace to tracking our personal health, the possibilities are endless.

4.3 Putting data at the heart

To deliver these experiences, organizations across India need to create an experience-focused mindset that is built on extracting value from data. Indian businesses are amongst the top to use that data to add value to its customers. India is advanced in its innovation around emerging technologies, such as artificial intelligence (AI), machine learning, and the Internet of Things. Indian business leaders have recognized that emerging technologies like AI need to be at the heart of their transformation, and around 8 in 10 Indian leaders have predicted that they will be using AI to pre-empt customer demands within five years.

As organizations derive more value from data – with those insights driving faster innovation and more efficient business processes – exciting times lie ahead for the technology sector. Digital transformation will use data and emerging technologies to create powerful new experiences and will strengthen businesses and allow them to remain competitive.

5. INNOVATIONS IN IMMERSIVE TECHNOLOGY

Immersive technology is booming in various sectors, from retail to health and the workplace. The UK government is investing 33 million GBP (1 Pound sterling equals 94.19 Indian Rupee) to support immersive technology research. A virtual education platform for learner drivers and VR therapy programmes to aid stroke victims and acrophobia sufferers.

5.1 AR HOLOGRAM APP

It is used to shoppers allows to view products in 3D.The app will be accessible through a smartphone or tablet with minimal data usage, turnaround time and processing power. AR's application in this industry could change the way consumers shop in the era of online shopping.

5.1.1 Software:

- PC Software
- Power Director
- Power DVD
- Screen Recorder
- You Cam
- Power2Go.

5.2 Virtual Business Card:

To create business card by using VR tool. For many people, business cards have become obsolete, supplanted by email, messaging and smart phones. However, a new augmented reality business card may help make them relevant again. It allows people to take a regular business card and add on AR objects, web pages, and contact information. As the card moves in the real world, the virtual elements move with it. The AR add-ons extend the limited information that will fit on a business card. Unlike a physical business card, Falmer's AR card needs a phone equipped with the card read app in order to view the extra AR features.

5.3 Lightweight Haptic Gloves:

It is used to user can 'touch' VR objects. As the business of virtual reality heats up, engineers and software developers have been working on technology that will allow users to touch virtual objects while also feeling like they are touching, grasping and manipulating real objects. When worn in sync with a suitable virtual reality programme, the haptic glove is supposed to react like a real glove that's touching real objects. Designed primarily for scientists, the glove is named DextrES. The glove is capable of generating up to 40 newton's of holding force on each finger. The system weighs less than 8 grams per finger and could be run on just a small battery. The glove, called DextrES, is made of nylon and thin elastic metal strips, separated by a thin insulator. When the user's fingers come into contact with a virtual object, the controller applies a voltage difference between the metal strips. Electrostatic attraction creates a braking force between the strips, which prevents the fingers from moving past a certain point. This gives wearers the impression that there is something in their hand. When the object is 'released' the voltage is removed. The metal strips will then separate and the user can once again move their fingers freely.

5.4 Smart Glasses:

This smart glasses project holograms only the wearer can see. The glasses, called Focals, use holographic display technology to overlay images onto the real world. One arm of the glasses contain a tiny projector. Images from the projector reflect off an element on the inside of the lens to focus the light back to the wearers' eye. The results are holographic images that only the wearer can see. A ring called the Loop controls the glasses and is worn on the index finger so the attached joystick can be controlled with the thumb. The Focal smart glasses join a host of other smart glasses projects, such as caption glasses that sync with theatre shows and lightweight AR glasses designed for the mass market. Every pair of Focals is custom-fitted to the wearers' head. To achieve this, North is setting up retail stores in Brooklyn and Toronto. The glasses are also compatible with Alexa. Users can ask Alexa questions and see the answer pop up in front of them. North is also working on integrating Focals into apps such as Uber and Google Maps. The glasses can also be used as normal spectacles for prescription users.

5.5 VR Suit:

It is created full-body VR suit. The seed-funded project's goal for virtual reality is to fully simulate the public's perception of life. Out of all the senses,

only vision and sound are on the necessary level according to the company. The third and most important sense, touch, is still under developed. VR to reach its full potential, technology companies have to consider touch in their offerings. The company's goal is therefore to create a full-body lightweight wearable haptic device to allow for active manipulation and interaction in virtual environments. Early stage businesses developing innovative and commercially focused applications of AR and VR. The program gives companies technical and business mentorship and access to state-of-the-art facilities. It concludes thereafter with the opportunity to pitch their ideas at an exclusive investor showcase. A haptic VR glove has transformed the lives of blind people by offering them the opportunity to experience art. Applications such as this exemplify how VR is so much more than a gaming tool. In a different industry, AR is being used for the fight against plastic pollution as a tool to advice customers on product in a zero-packaging store. These varied examples demonstrate the potential of state-of-the-art technology in multiple capacities.

5.6 VR App:

It is aids learner drivers. Virtual Reality (VR) has weaved itself into every industry, proving itself to be useful in different capacities. Spring wise has covered the technology in length, analyzing the multiple ways in which it is changing businesses and sometimes even people's lives. One VR tool used in hospitals shows pediatric patients a 3D tour of their body, helping alleviate fear in a stressful situation. Elsewhere, the technology has proved useful in business settings by connecting employees via a teleconference to analyses data. Who knows how VR will expand next, but it is inevitable the system will reach the forefront of many business strategies. The Drive VR app was developed to help people learning to drive to gain experience. It can also be used to boost confidence in everyday driving skills such as observation tasks. It is hoped the platform will help learners gain experience and knowledge for their theory test through to the practical exam. App users will get the chance to practice spotting hazards, checking blind spots and mirrors and looking out of windows at the right time. Observation skills are tracked, so learners can keep track of their progress rates.

5.7 Stroke Victims to Aid Recovery:

VR could trick for stroke victims to aid recovery. We have seen how virtual reality can be used for allowing collaborative design and controlling robots remotely. Now, researchers believe that VR may be able to help stroke victims regain lost motor function. Research conducted by University of Barcelona professor Mel Slater and Stanford professor Jeremy Bailenson has demonstrated that virtual experiences can lead to changes in perceptions of oneself and others. For example, having an avatar of a child in a virtual reality environment can encourage people to exhibit more childlike behaviour in the 'real world'. Inspired by this work, researchers at the University of Southern California are now examining how virtual reality could promote brain plasticity and recovery. The team hopes that the resulting visual feedback through a VR headset could help strengthen neural pathways from the damaged motor cortex to the impaired arm, leading to real improvements in mobility. The research group recently finished testing the prototype with 22 healthy older adults, who provided data on what the brain and muscle signals look like when they move. They are now starting to test with stroke patients in a controlled lab setting.

5.8 3D Data:

VR used to analyze 3D data. The technology helps users navigate data by grabbing and interacting with it. Lume, a VR technology startup, was chosen as part of Digital Catapult's Augmenter 2018 intake. Augmenter is a 10-week programme supporting early stage businesses developing innovative and commercially focused applications of AR and VR. The programme gives companies technical and business mentorship and access to state-of-the-art facilities. This culminates with the opportunity to pitch their ideas confidently at an exclusive investor showcase. VR offers the opportunity to view age-old things in a different way. The technology is opening doors to helping to educate, inform and expand a person's experiences. Its capabilities are a far cry from its native video game roots. From an educational perspective, a VR app has been built to assist learner drivers with boosting their confidence and road awareness. This style of application could prove priceless in environments where young people struggle to consume information required for exams via books, for example.

5.9 Overcome Acrophobia:

VR therapy programme aims to reduce the fear of heights with computer-generated virtual coach. Utilizing the way people think is one of the most astonishing ways in which technology can function. From block chain technology tackling extremist views to children learning about anti-discrimination via an online soccer game, its use is seemingly endless. The team has developed a VR programme in which psychological therapy is delivered by a computer-generated virtual coach. Users receive personalized treatment, therefore being able to interact with the virtual coach using voice recognition technology. A random selection of one hundred people with a fear of heights received either the VR therapy or no treatment. Participants had, on average, lived with a fear of heights for 30 years. Those who received the therapy spent an average of two hours in VR over five treatment sessions. All participants in the VR group showed a reduction in fear of heights, with the average reduction being 68 percent. Furthermore, half of the participants in the VR group had a reduction in fear of heights by over three quarters.

5.10 Zero Packages:

Zero packaging store uses AR to provide product info. A retail store has launched packaging free products, where customers can use a recognition app to retrieve AR product information.

Plastic waste from retail packaging is a big problem. Discarded containers are finding their way into islands of ocean plastic and causing havoc to marine wildlife and beyond. We're seeing attempts to clean up our oceans and to recycle the plastic into functional, durable products. Lush, however, is tackling the problem at the source by targeting one of the main issues of plastic waste: cosmetic packaging. However, when packaging is removed, a crucial source of product information is lost. This is where the app, also developed by the R&D team, comes in. The store has partnered with Fairphone to provide in-store smart devices loaded with the app. Customers can point the device's camera at the product they're interested in. The software will recognize the product and display an AR (augmented reality) information page about the product's ingredients.

6. CONCLUSION

Immersive VR can be defined by its technology and its effects. Its primary effect is to place a person into a simulated environment that looks and feels to some degree like the real world. In immersive VR, simulated objects appear solid and have an egocentric location much like real objects in the real world. They can be picked up, examined from all sides, navigated around, heard, smelled, touched, hefted, and explored in many sensory ways. The objects can also be autonomous (especially if they are other people) and interact with the virtual voyager, or respond to voice commands. The fundamental limitation to all these effects is in the computational technology that supports them. It uses many hardwares and soft wares which helps it to function efficiently. Its applications are covered almost in all fields. Finally, Immersive Technology is an emerging technology

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